

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Summer Supplementary Examination – 2023**

**Course: B. Tech.**

**Branch : AI&DS and Allied**

**Semester : III**

**Subject Code & Name: BTESC304 & Computer Architecture & Operating Systems**

**Max Marks: 60**

**Date:18/08/2023**

**Duration: 3 Hr.**

***Instructions to the Students:***

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

|                                                         | (Level/CO) | Marks     |
|---------------------------------------------------------|------------|-----------|
| <b>Q.1 Solve Any Two of the following.</b>              |            | <b>12</b> |
| A) Explain briefly about computer fundamental system?   | L1/CO1     | 6         |
| B) Explain instruction set Architecture? Give examples. | L1/CO1     | 6         |
| C) Give the difference between RISC and CISC.           | L1/CO1     | 6         |
| <b>Q.2 Solve Any Two of the following.</b>              |            | <b>12</b> |
| A) Explain Multistep Processing of a User Program.      | L2/CO2     | 6         |
| B) Explain Implementation of a Page Table.              | L2/CO2     | 6         |
| C) Write a short note on Virtual Memory.                | L2/CO2     | 6         |
| <b>Q.3 Solve Any Two of the following.</b>              |            | <b>12</b> |
| A) Explain Programmed I/O Mode in detail.               | L3/CO3     | 6         |
| B) Describe Direct Memory Access.                       | L3/CO3     | 6         |
| C) Write a short note on SCSI.                          | L3/CO3     | 6         |
| <b>Q.4 Solve Any Two of the following.</b>              |            | <b>12</b> |
| A) Explain Multiprogramming.                            | L4/CO4     | 6         |
| B) Describe Batch Processing.                           | L4/CO4     | 6         |
| C) Explain Distributed Systems.                         | L4/CO4     | 6         |
| <b>Q.5 Solve Any Two of the following.</b>              |            | <b>12</b> |
| A) Describe Reader –writer problem Problem.             | L5/CO5     | 6         |
| B) Explain Deadlock Characterization.                   | L5/CO5     | 6         |
| C) Describe Methods for Handling Deadlocks.             | L5/CO5     | 6         |

**\*\*\* End \*\*\***